

CEDRIC P.-E. MANOUAN

DEEP LEARNING ENGINEER - AI SPECIALIST

Contact : +225 07 59 20 15 98 | **Email:** manouancedric@gmail.com | **Linkedin:** [Cédric Pascal-Emmanuel Manouan](#)
Github: [dric2018](#) | **Website:** dric2018.github.io

EDUCATION

Master of Science <i>Information Technology - Applied Machine Learning</i> Carnegie Mellon University - College of Engineering, CMU-Africa	August 2021 – May 2023 Kigali, Rwanda
Master of Science (Semester 7) <i>Information Systems and Software engineering</i> Ecole Supérieure Africaine des TIC (ESATIC)	October 2019 – March 2020 Abidjan, Côte d'Ivoire
Bachelor of Science <i>Systems, Computer Networks and Telecommunications</i> Ecole Supérieure Africaine des TIC (ESATIC)	October 2016 – September 2019 Abidjan, Côte d'Ivoire

WORK EXPERIENCE

International Finance Corporation (IFC), World Bank Group <i>Project Coordinator – AI Associate Specialist (Short Term Consultant)</i>	September 2025 – June 2026 Abidjan, Côte d'Ivoire & Remote
- Coordinated internal and external dissemination of project outputs - Supported the preparation of the AI-related materials that shaped IFC's contribution to the GITEX Africa 2026 panel on AI, Science & Global Collaboration presented by our Division Director - Supported the WBG teams on AI compute projects	
— <i>Project Analyst – AI Associate Specialist (Short Term Consultant)</i>	April 2024 – September 2025 Abidjan, Côte d'Ivoire & Remote
- AI and Startup Ecosystem Policy Research (West and Central Africa) - Co-authored 3 AI landscape reports on startups and AI covering Cameroon, Côte d'Ivoire and Senegal - Contributed inputs on AI and digital skills to the preparation of World Bank's Digital Acceleration Program in Côte d'Ivoire (PADCI)	
Techclick, UNDP & Rwanda Ministry of ICT and Innovation (MINICT) <i>Consultant - AI Readiness Assessment (AIRA)</i>	October 2023 – March 2024 Kigali, Rwanda
Supported the UNDP team with the qualitative assessment of the AI landscape	
Carnegie Mellon University Africa <i>Consultant (AI Engineer)/Principal Research Associate, Wakanda AI, AFRETEC Network</i>	July 2024 – June 2025 (Remote) Kigali, Rwanda
- Drafted initial paperwork (proposal, budget, work plan); - Speech processing, Language modeling; Seamless communication; African languages	
— <i>Research Associate, AI & Robotics Lab</i>	May – December 2023 Kigali, Rwanda
Implementation of a Semantically-modulated Joint Episodic-Procedural (S-JEP) Working Associative Memory	
— <i>Graduate Research and Teaching Assistant</i>	Fall 2022 & Spring 2023 Pittsburgh, PA / Kigali, Rwanda
11785 - Introduction to Deep Learning 04800-AB Machine Learning for Earth Observation	
Ivorian Electricity Company (CIE) <i>AI Analyst Intern</i>	June 2022 – July 2022 Abidjan, Côte d'Ivoire
Zindi <i>Software Engineering/Data science Intern</i>	April 2021 – July 2021 Remote
PSNDEA <i>Software Developer Intern</i>	May 2019 – July 2019 Abidjan, Côte d'Ivoire
Project of Digital Solutions for the opening up of Rural Areas and e-Agriculture	

SKILLS

Programming (Python/C++) & Software/AI Engineering: Data structures and Algorithms; Deep Learning (PyTorch/Lightning; Large Language Models; Agentic and local AI); API development (FastAPI/NodeJS), Amazon Web Services (AWS/EC2); CI/CD; Docker.

High-performance computing (HPC): Distributed job scheduling on compute (GPU) clusters, resource management, and batch script development using SLURM.

Consulting: Information Technology (IT), AI (Governance; Readiness Assessment; Engineering; Project Management).

FEATURED PROJECTS

Take The Train - Africa at the Crossroad of Modern AI | *Independent Research* March 2026
Paper link: <https://arxiv.org/abs/2603.21795>

- Led research on Africa’s AI landscape, identifying critical barriers categorized into infrastructure, accessibility, governance, and human capital challenges, and proposed a framework built around the “right enablers” for sustainable AI adoption across Sub-Saharan Africa.
- Designed the Africa AI Compute Tracker (ACT), an interactive map consolidating HPC availability data across Africa by September 2025. to foster transparency and regional collaboration.

Wakanda AI - Advanced Speech Technologies for African Languages (2024-2027) | *CMU-Africa*
Consultant (AI Engineer), Principal Research Associate
\$250,000 Research grant to Advance the Inclusive Digital Transformation of Africa, funded by the Mastercard Foundation through the AFRETEC Network.

- PI: Prof. Bhiksha Raj, Language Technologies Institute, Carnegie Mellon University

NLP & Robotics - Robotics Transformer for CRAM (RT1-CRAM) | *CMU-Africa* Fall 2023
PyTorch re-implementation of RT-1 for the generation of robot actions represented as sequences of text tokens composed of the parameters to a CRAM generalized action plan.
Project link: [S-JEP \(RT1-CRAM\)](#)

HONORS AND AWARDS

Mastercard Foundation 2021 – 2023
Full scholarship; completed the Marketplace Readiness Leadership Training Programme Kigali, Rwanda

ESATIC 2017 & 2018
2x First Prize, Technovore Hackathon (Software development, AI/ML) Abidjan, Côte d’Ivoire

LEADERSHIP & COMMUNITY INVOLVEMENT

Advanced Computing and Data Resource Program (ACCESS) 2023 - 2025
Member, Researcher Advisory Committee Remote, USA

SAIEE Africa Research Journal (incorporating the SAIEE Transactions) Spring 2024
Reviewer Remote

International Conference on Learning Representations (ICLR) Spring 2023
Student Volunteer Rwanda, Kigali

REFERENCES

Ms. Meritxell Martinez	Senior Operations Officer, IFC	mmartinez1@ifc.org
Mr. Paul Nguyen	Digital Specialist, The World Bank	pnguyen19@worldbank.org
Prof. Bhiksha Raj	Professor, Language Technologies Institute, CMU	bhiksha@cs.cmu.edu
Prof. David Vernon	Emeritus Research Professor, CMU-Africa	david@vernon.eu
Prof. Mark Kryder	Emeritus ECE Professor, CMU	mk6x@andrew.cmu.edu